

QSort Activity: Gauging perspectives on the effectiveness and feasibility of different policies/strategies for supporting EV evacuation during extreme events

- Please introduce your name in the green slide
- The subsequent slides contain the following:
 - Summary of policies/strategies you will be assessing
 - QSort tables to guide your categorization of each strategy relative to each other
 - Additional questions for you to reflect upon after completing the sorting activity
- Please let us know if you have any questions or would benefit from some additional clarification

INTRODUCE YOUR NAME HERE

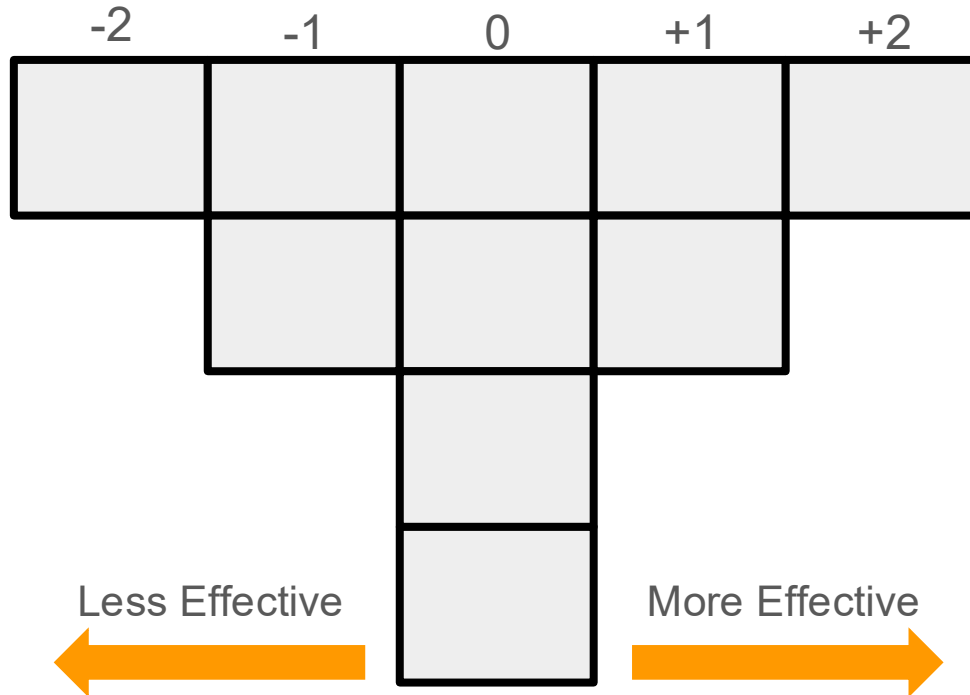
Summary Table of Strategies

Strategy #	Strategy	Description
1	Backup energy – fossil fuels	Require public charging stations to have on-site backup energy to maintain operability during power outages. The backup power would come from diesel generators (or similar)
2	Backup energy - microgrid	Require public charging stations to have on-site backup energy to maintain operability during power outages. The backup power would come from microgrids consisting of some combination of solar/wind energy and battery energy storage
3	Free/open charging	During emergency declarations, public charging stations would be “unlocked” so that people can essentially drive up and immediately start charging without having to pay or interface with any kind of app or point of service system. Charging companies could be reimbursed by federal/state governments
4	Mobile chargers at origin	Dispatch mobile chargers (of any type) to evacuation zones prior to the event to increase the initial state of charge and reduce the need for charging along the evacuation route
5	Mobile chargers to supplement existing charging locations	Dispatch mobile charges (of any type) to sites with existing stationery charging infrastructure. The mobile chargers would act as supplemental resources to the existing charging infrastructure

Summary Table of Strategies (Continued)

Strategy #	Strategy	Description
6	Mobile chargers at designated locations along evacuation route	Dispatch mobile chargers (of any type) to designated locations (e.g., rest stops, shopping centers, etc.) along evacuation routes that have minimal existing charging infrastructure
7	Minimum “charge reserve” required on all EVs	On all EVs, require a certain % of battery charge to be kept in reserve that can be unlocked during an emergency declaration
8	Bussing/shuttling people out of evacuation zones	Use buses or shuttles (or any type) to facilitate the evacuation of people rather than relying on personal vehicles. This could reduce congestion on the road, as well as demand at downstream charging stations
9	Staggered departures from evacuation zones	Specify time periods when different locations can evacuate. This could reduce congestion on the road, as well as demand at downstream charging stations
10	Minimum “spinning reserves” required in the power grid	Require utility companies and electricity grid operators to keep a certain amount of electricity generation capacity on “standby”. This can support a rapid uptick in charging demand prior to and during an evacuation. It also provides some “backup” capacity in reduced generation from solar/wind

QSort 1: Which strategies/policies do you feel will be most/least effective at supporting evacuation of EVs during an extreme event?



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| Strategy 1:
Backup
Energy
(fossil fuels) | Strategy 6:
Mobile
chargers
along route |
| Strategy 2:
Backup
Energy
(microgrid) | Strategy 7:
Minimum EV
charge
"reserve" |
| Strategy 3:
Free/open
charging | Strategy 8:
Bussing/
shuttling
people |
| Strategy 4:
Mobile
chargers at
origin | Strategy 9:
Staggered
departures |
| Strategy 5:
Mobile
chargers at
existing spots | Strategy 10:
Minimum grid
"spinning
reserves" |

QSort 1 - Reflection Questions

- For the two strategies on the far-left (-2) and far-right (+2) of your table, why did you choose those as the least/most effective strategy? Do you have any real-world examples demonstrating your reasoning?
 - [\[Insert response here\]](#)
- Which of the strategies were most difficult to categorize? Why?
 - [\[Insert response here\]](#)
- What criteria (if any) did you use to sort the strategies? How would you describe/define a “successful” evacuation?
 - [\[Insert response here\]](#)
- Are there any additional strategies/policies not included in this exercise that you think might be relevant or helpful for ZEV evacuations?
 - [\[Insert response here\]](#)

{If needed, additional space is available on next slide}

Additional space for response QSort 1 reflection questions